INCH-POUND

MS25030P
03 January 2023
SUPERSEDING
MS25030N
27 November 2003

DETAIL SPECIFICATION SHEET

RELAY, 50 AMP, 2 PDT (N.O.), TYPE II, NON-HERMETICALLY SEALED, MECHANICALLY INTERLOCKED

Reactivated after 03 January 2023 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

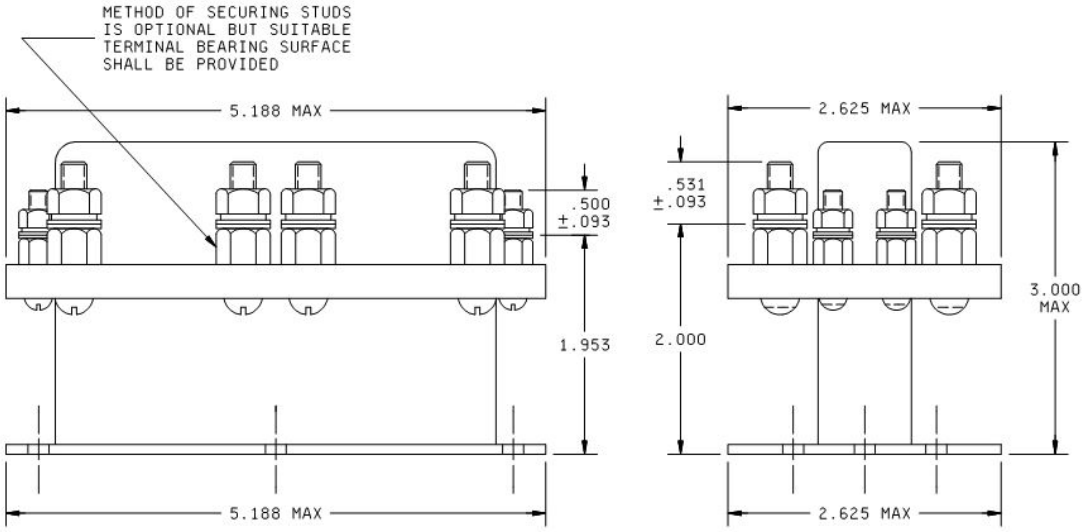
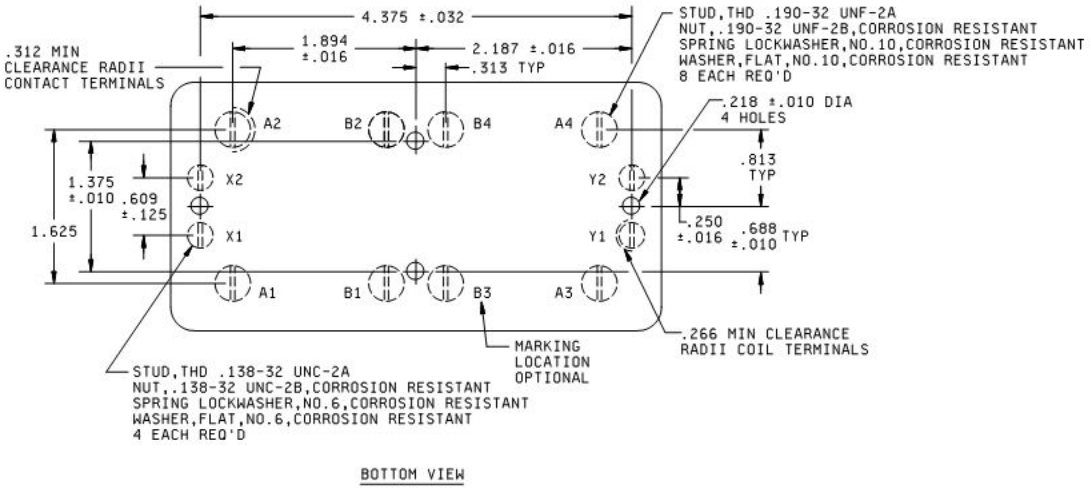


FIGURE 1. Dimensions and configuration.

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Inches	mm	Inches	mm	Inches	mm
.010	0.25	.313	7.95	2.000	50.80
.093	2.36	.500	12.70	2.281	57.94
.125	3.18	.531	13.49	2.625	66.68
.138	3.51	.609	15.47	3.000	76.20
.190	4.83	.688	17.48	4.562	115.88
.218	5.54	.813	20.65	5.188	131.78
.266	6.76	1.375	34.93		
.305	7.75	1.625	41.28		
.312	7.93	1.938	49.23		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.062 (1.57 mm).
4. Coil and auxiliary terminals may use additional flat washer for terminal seat.
5. Mechanical linkage shall be provided to prevent both relays from being closed simultaneously.
6. Referenced Government documents of the issue listed in that issue of the Acquisition Streamlining and Standardization Information System (ASSIST) specified in the solicitation forms a part of this specification sheet to the extent specified herein.
7. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENTS:

Dimensions and configurations: See figure 1.

Dash numbers and general characteristics: See table I.

Contact data:

Load ratings: See table II.

Maximum contact drop, initial: 0.150 V.

After life test: 0.175 V.

Overload current (NO): 400 amperes.

Rupture current (NO): 500 amperes.

Coil data: See table III.

Duty rating: continuous.

RFI specification: MIL-STD-461 (applicable to coil circuits of ac operated relays).

Electrical data:

Minimum insulation resistance:

Initial: 100 megohms.

After life or environmental test: 50 megohms.

Dielectric strength:

Sea level, 2-5 seconds:

	Initial		After life tests	
	28 V dc	115 V ac	28 V dc	115 V ac
Coil to case	1,250	N/A	1,000	N/A
Aux contacts	1,250	N/A	1,000	N/A
All other points	1,250	1,500	1,000	1,125

Altitude, 1 minute:

	28 V dc	115 V ac
Coil to case	500 V rms	N/A
Aux contacts	500 V rms	N/A
All other points	500 V rms	500

Environmental characteristics:

Temperature range -55° to +71°C.

Maximum altitude rating: 50,000 feet.

Shock g-level: 25 g's.

Duration: 6-9 ms.

Max duration contact opening: 2 ms.

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Vibration - sinusoidal: See table IV.

Vibration - random: Not applicable.

Acceleration: 10 g's.

Part or Identifying Number (PIN): MS25030- (plus applicable dash number from table I).

Qualification by similarity: See MIL-PRF-6106.

Assurance provisions: Group B and group C testing are not required. In the event of a change in the design or construction of the part, the manufacturer shall notify the qualifying activity and shall impose additional testing requirements as necessary.

Supersession data: See table V.

TABLE I. Dash numbers and general characteristics. 1/ 2/

PIN	Type	Coil type	Terminal type	Mounting or mating socket	Auxiliary contacts	Maximum weight in pounds 1/
MS25030-D1B	II	dc	Stud	Plate	N/A	1.7

TABLE II. Rated contact load (amperes per pole) (case grounded). 1/

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase				See notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	
Resistive	50	50				50			5					
Inductive	10	50							5					
Motor	50	50				35								
Lamp									.75					
Transfer load														2/
Mechanical life reduced current	100	12.5				12.5			1.25					
Mixed loads	50	5				10								

1/ Absence of value indicates relay is not applicable to this specification sheet.
2/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplies at rating indicated.

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TABLE III. Operating characteristics.

PIN MS25030-	Coil data											Time – milliseconds max 2/					
	Coil	Rated			Max		1/ Max pick-up voltage			Hold volt-age 2/	Drop out volt-age 2/	Oper -ate 3/	Re-lease 4/	Bounce			
		Volts 1/	Freq Hz	Ω Res +15% -10	Volts	Amp	Nor-mal 2/	High temp test	Cont cur-rent test					Main		Aux	
														NO	NC	NO	NC
D1B	X1, X2 Y1, Y2	28	dc	66	29	.5	18	21	22.5	7.0	1.5	25	15	5	N/A	N/A	N/A

1/ Over the temperature range.
2/ With rated coil voltage.

TABLE IV. Vibration levels (sinusoidal).

Dash number	5-10 Hz	10-55 Hz	55-250 Hz	250-500 Hz	500-1,500 Hz
D1B	.08 DA	.06 DA	2 g's	2 g's	N/A

TABLE V. Supersession data.

Superseded PIN MS25030-	Replacement PIN MS25030-
2	D1B
3	D1B
D1A	D1B

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. In addition to MIL-PRF-6106, this document references the following:

MIL-STD-461

Custodians:
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-2023-009)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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